Commonality in Liquidity- New Evidence from National Stock Exchange, India[1]

Gaurav Kumar

(Corresponding Author)

Indian Institute of Technology (IIT), Kharagpur, India

And

University College Dublin (UCD), Dublin, Ireland

Email: gauravkumar@iitkgp.ac.in, gaurav.kumar@ucd.ie, Mobile: +353-899875952

Arun Kumar Misra

Indian Institute of Technology (IIT), Kharagpur, India

Email: arunmisra@vgsom.iitkgp.ernet.in, Mobile: +91-9832908136

[1] This article is based on a part of Kumar's PhD dissertation at IIT Kharagpur, India.

Commonality in Liquidity- New Evidence from National Stock Exchange, India

ABSTRACT

This article investigates hypothesis related to commonality in liquidity on National Stock Exchange (NSE), India using high frequency limit order book data. The empirical analysis shows that individual stock liquidity based on various spread and depth measures co-moves with market liquidity and industry liquidity. Market-wide commonality is found stronger than Industry-wide commonality on majority of the liquidity measures. The study captures the asymmetric behaviour of commonality in up and down markets. Among four key sectors of the economy *viz.* Consumer Goods, Financial Services, Manufacturing, and Infrastructure –the strength of commonality is higher in Manufacturing sector. The study develops long run systematic liquidity measure to investigate commonality sources in long run and compares it with the short run. The results of this study can be used by traders in devising strategies, exchanges in designing trading platforms, and fund managers in engineering financial instruments.

KEYWORDS: Liquidity Commonality; Market Microstructure; Indian Stock Market; NSE

JEL CLASSIFICATION: G11; G12; G15

1. Introduction

Commonality refers to the co-movement phenomenon between individual stock liquidity and market liquidity. Earlier strand of literature on market microstructure consider liquidity to be an attribute of a single asset {Amihud and Mendelson (1980), Kyle (1985),Admati and Pfeiderer (1988), Grossman and Miller (1988)}. However, recent studies provides evidence about the systematic component of liquidity (Chordia, Roll and Subrahmanyam, 2000; Huberman and Halka, 2001). Chordia *et al*. (2000) provides evidence that individual stock's liquidity changes are positively and significantly related to changes in market liquidity while Hasbrouck and Seppi (2001) found no such evidence. The empirical manifestation of commonality is the co-movement between variations in individual stock's liquidity and variations in market and industry wide liquidity (Chordia *et al*., 2000). Commonality is a systematic risk which cannot be diversified or hedged by the traders. As, the asset is more sensitive to the time variation in commonality of liquidity it results in to greater expected returns {Giouvris (2003); Acharya and Pederson (2005)}.

National Stock Exchange (here after NSE), India is an order driven markets where as NYSE is a quote driven market. The structural differences between the two exchanges could result in differences in strength of commonality on the basis of depth and spread measures (Brockman and Chung, 2002). Quote driven markets are characterized by presence of market makers and dealers. It would be interesting to test the model of Chordia *et al.* (2000) in a different Order driven market structure like India, where these players are absent to supply or demand liquidity.

The motivation behind studying commonality is that it will help regulators and researchers gain insights of the factors that affect the sensitivity of the liquidity in Indian market and implement policies that might prevent turmoil from liquidity shocks. The analysis of this study has important implications for the portfolio managers as they are not only concerned with the liquidity of single asset but how asset's liquidity co-moves with the liquidity of other assets. Liquidity measured by quotes and quantities beyond best prices contains information required by the investors to position large trade {Domowitz, Hansch and Wang, 2005; Kempf and Mayston, 2006}. However, many researchers based commonality evidence on the basis of best bids and offers {Chordia et al., 2000; Choe and Yang, 2010}.The strength of this study lies in providing commonality evidence on the basis of quotes beyond best bids and asks in the limit order book.

Studying liquidity movements on NSE systematic is prominent for Foreign Institutional Investors (FII's) and domestic investors, as NSE is the fifth largest stock exchange of Asia with a total market capitalization of $1.5 Trillion (see Appendix II). India ranks among top five fastest growing economy of the world as per the growth rate projections by the World Bank. It is backed by ample resources, strong macroeconomic fundamentals, favorable demographics, robust domestic demand and a proactive central government. However, only a very low proportion of the household savings of Indians are invested in the domestic stock market. But with a stable financial market and high GDP growth rate, we might see more money flowing in to capital markets. It's the right time for foreign portfolio investors to seriously think about investing the India bandwagon. At this stage of growth, it is necessary to assess the level of systematic liquidity in Indian stock market in order to establish its longer term role in the process of economic development. All in all, Indian stock market is good platform to test commonality hypothesis across market, industry, in up and down market and over the long run.

Driven by such research and investment motivations, we investigate the issue of commonality along five measures of liquidity in an open electronic limit order book market, which is the market design of NSE. In particular, the analysis is focused on Midcap stocks listed on NSE, and employs a 4-month high frequency data set that contains the entire trading history (orders and trades). The study utilized the time-series regression model of Chordia *et al.* (2000) and Faff *et al.* (2002) to capture liquidity co-movements in long run and short run respectively. We assess the magnitude of commonality on individual stocks liquidity in up and down markets. This makes the research design more comprehensive and as well as informative. The study also tested commonality hypothesis across four key sectors of the Indian economy, thus making findings of the study more robust and valid. The existence of commonality as a non-diversifiable priced risk implies that the investors would demand higher expected returns from stocks with higher sensitivity to market-wide liquidity shocks and industry-wide liquidity shocks (Chordia *et al.*, 2000).

Rest of the article is organised as follows. A brief summary of the literature is discussed in section 2. Section 3 outlines institutional setting of NSE. Section 4 describes the variables and presents summary statistics of the data. Section 5 presents research methodology and discusses empirical results in light of the results reported in existing literature. Section 6 summarises the major findings of the article.

2. Theoretical Rationale

Events such as market Crash of October 1987 and East Asian Financial Crises of 1997 indicates that rise and fall of individual stock liquidity being influenced by market liquidity. This co-movement between variations in individual stock liquidity and variations in market and industry wide liquidity is termed as Commonality (Chordia *et al.*, 2000). In recent years, liquidity commonality has drawn attention of the scholars because of its prominent implications for traders, stock exchanges, regulators and firms. Many studies document commonality phenomenon in quote-driven markets (Chordia *et al.*, 2000; Huberman and Halka, 2001; Dicle and Mukherjee, 2011). Coughener and Saad (2004) argued that specialist firms induce commonality in quote driven markets such as NYSE. Specialists within the same firm share information and capital. This causes liquidity of an individual stock co-moving with the liquidity of other stocks handed by the same specialist firm. However, most of the Asian markets *viz.* China, India *etc.* are order-driven. The absence of specialist firms suggests that information and capital sharing are less relevant in order-driven markets, as compared to the markets that have specialists, such as NYSE or Nasdaq. Chordia *et al.* (2000) empirically studied common underlying determinants of time series movements in liquidity, known as commonality. Their study modelled daily percentage changes in an individual stock's liquidity on percentage changes in the equal-weighted liquidity measures for all stocks within the market and within the same sector. This is evidence indicating that inventory risks and asymmetric information causes inter-temporal changes in individual stock liquidity.

Huberman and Halka (2001) statistically investigated 60 NYSE stocks and reports evidence of systematic component of liquidity. Presence and effect of noise traders is conjectured for temporal variations in liquidity. In contrast, Hasbrouck and Seppi (2001) reports weak commonality on spread and depth based measures as judged by first principal components on 30 Dow stocks. Brockman and Chung (2002) investigated commonality on order driven Stock Exchange of Hong Kong. The study finds that commonality in liquidity is perseverant across all size based portfolios. Market-wide component of the systematic liquidity is stronger than Industry-wide counterparts. Commonality on the basis of spreads is found to be more pronounced in middle sized firms. These results are in contrast with Chordia, Roll and Subrahmanyam (2000), reporting that large firms are more susceptible to commonality. Brockman and Chung (2002) argued that commonality is more pronounced in order driven systems because of absence of market makers who have an obligation to maintain orderly and fair market. Pukthuanthong-Le and Visaltanchoti (2009) found evidence of strong market-wide commonality on the basis of liquidity beyond best bid and ask on order driven Stock Exchange of Thailand (SET). Industry-wide commonality is found to be stronger than market-wide commonality. Small firms are more susceptible to market-wide liquidity based on spread based measures while large firms are more susceptible to market-wide liquidity based on depth based measures.

Narayan *et al.* (2015) examined four commonality hypotheses on Shanghai and Shenzhen stock exchanges. First, market-wide liquidity is variable causing variation in liquidity of individual stocks. Second, size of the firm is not a determinant of commonality on Chinese stock exchanges. This is different from the existing literature which says size effects commonality {Brockman and Chung, 2002; Sujoto *et al*., 2005}. Third, sector specific liquidity has a greater influence on liquidity of individual stocks in comparison to market-wide liquidity. Out of the sectors *viz*. Industrial, Resources and Financial, it is the Industrial sector which is highly correlated with individual stock liquidity. Fourth, commonality in liquidity varies in up and down markets.

Commonality is found stronger during down period then up period as investors are more concerned of macroeconomic news in comparison to firm performance.

Tayah *et al.* (2015) found market-wide commonality on Amman Stock Exchange stronger than on Hong Kong (Brockman and Chung, 2002) and Australia (Fabre and Frino, 2004). However, Industry-wide commonality is not found significant on Amman Stock Exchange. The results are consistent with the study of Galariotis and Giouvris (2007), and Fabre and Frino (2004) but inconsistent with the study of Chordia *et al.* (2000), Brockman and Chung (2002) and Pukthuangthong-Le and Visaltanachoti (2009). Small stocks are found to be more susceptible to commonality in liquidity on the basis of effective spread, quoted spread and proportional quote spread. Papachristou *et al.* (2016) examined commonality existence on thinly traded and pure order driven Athens Stock Exchange (ASE). Commonality is found to be low using bid-ask spread. However, it increases on moving deeper in the order book. The study argued that investor's portfolios are more susceptible to commonality as larger transactions consume liquidity beyond the best quotes on thinly traded exchange. This study also reports that lead and lag liquidity commonality is more pronounced indicating strong evidence of asynchronous liquidity commonality.

Some literature explores the potential sources of liquidity commonality. Chordia, Roll and Subrahmanyam (2001) identified macroeconomic news, calendar regularities, short term interest rates, long term spreads, market returns and volatility to be determinant of trading activity and liquidity. These determinants explain 18% to 33% changes in liquidity and trading activity. Brockman and Chung (2006) proposed index inclusion hypothesis according to which index inclusion is a source of commonality in liquidity. This study empirically confirmed that the stock that belongs to equity index has a greater exposure to communality then the one that is not a part of any equity index. Goyal *et al.* (2014) considers four time instances during the trading day to study commonality on NSE. The study reported size effects and dominance of industry-wide commonality over market-wide commonality.

Understanding commonality in long run helps in tracking liquidity movements because of noise traders and informed traders. Faff *et al.* (2008) develops long run commonality beta (β_L) by augmenting the model developed by Chordia *et al.* (2000). The study reports large proportion of significant and positive β_L which indicates support for systematic liquidity in the long run. However, this beta (β_L) is found to be smaller than short-run impact beta indicating co-movement of liquidity is caused by noise traders activity.

Pricing of commonality is an important issue for market participants as it is systematic risk. Acharya and Pederson (2005) made a thorough examination on the relationship between stock returns and various channels of systematic liquidity risk. Anderson *et al.* (2013) reported statistical significance of return premium for the commonality risk on NYSE listed stocks.

Commonality in liquidity is an area of market microstructure that has not been addressed thoroughly in the Indian Stock Market. Kumar and Shah (2012) made limited study of commonality on NSE, by using price impact as the only liquidity proxy and four limit order snapshots per day. The study reported economic significance of the existence of commonality on the basis of firm sizes, Calendar effects (day, month, year), and asymmetric commonality exhibited during bull and bear phases of the market. Krishnan and Mishra (2015) also tried to detect commonality using principal components analysis (PCA) on 20 NIFTY stocks over 15 minute interval data. Only 18% of the total variation is explained by first eigen value, indicating very weak commonality across the measures. Kempf and Mayston (2006) argued that weak commonality is because of idiosyncratic variation and noise in the best quotes.

Studies that investigate the commonality in emerging Asian markets like India are very few. The reason for the lack of studies is mainly because the transaction and limit order book data are typically difficult to obtain for non-US exchanges. Kumar and Shah (2012); and Goyal *et al.* (2014) reported strong presence of commonality. However, the analysis of Krishnan and Mishra (2013) provides weak evidence of commonality using principal component analysis (PCA).
Therefore, no consensus has been developed among researchers on the evidence and strength of commonality on Indian stock markets. Furthermore, there are drawbacks associated with the previous investigations. The studies of Kumar and Shah (2012); and Goyal et al. (2014) analyzed limit order book at particular snapshots of time and considered only best bid and best ask quotes. Capturing trade and quote data at particular snapshots can lead to missing data problem as limit orders may not be available at that time. These studies fail to extend their analysis beyond the prevailing best quotes. Best quotes are noisy and may not be well-suited for liquidity commonality analysis (Pukthuanthong-Le and Visaltanchoti, 2009). Also, liquidity measured beyond best quotes would affect the behavior of market participants and the price discovery process, which further affects the commonality in liquidity. Therefore, researchers should consider complete order book comprising of quotes and quantities beyond best prices (Domowitz, Hansch and Wang, 2005; Kempf and Mayston, 2006).

Another significant gap in the existing literature is that most of empirical studies is focused around flagship index stocks {Hasbrouck and Seppi (2001) on 30 DOW stocks; Papachristou *et al*. (2016) on ATHEX-20 blue chip stocks; Krishnan and Mishra (2013) on 20 NIFTY stocks}. Flagship index stocks are more liquid as compared with the other listed stocks. In contrast to the existing literature, our study is limited to Midcap stocks which are relatively less liquid. These stocks are highly attractive for investment, and having potential for becoming Large-Cap.

In this paper, we attempt to fill these research gaps for the Indian stock market and extend the notion of commonality in liquidity to entire the limit order book. Our study empirically examines the applicability of following research questions and hypotheses for the Midcap stocks listed on NSE using high frequency data:

a) Market-wide liquidity drives liquidity of individual stocks *i.e.* is there commonality in liquidity of the stocks listed on NSE?
b) What is the relative strength of market-wide commonality and industry-wide commonality?
c) Is commonality varies with the Industry?
d) What is the relative strength of commonality in up and down markets?
e) Is there evidence of the long run commonality in liquidity?

3. Market Design: Indian Stock Markets (NSE)

Market design is key aspect of market microstructures. Market conditions are influence by it which in turn impacts the price formation process. Markets around the world are designed on the basis of order execution system i.e. matching buyers to sellers. NSE is one the leading exchanges in the world. By 2020, the size of the capital markets in India will become USD 8 trillion[2]. NSE is pure order driven market, in which the buyers are looking for the lowest available prices and sellers are looking for the highest available prices. The phenomenon is known as price discovery as it reveals the prices that best match to buyers and sellers. The order precedence rule for the price discovery is determined on the basis of price-time priority. The orders are first sequenced on the basis of best price and within best price, time. Limit orders get priority over market orders at the time of execution of trades. Unlike NASDAQ (USA), there is no market maker on NSE.

[2]FICCI and McKinsey knowledge paper. Available at: http://ficci.in/spdocument/20191/FICCI-CAPAM-2012-Knowledge-paper.pdf

Market timings of NSE are from 9:00 to 15:30 hours. The first 15 minutes is known as pre-open session (call auction) is held to contain the volatility and price discovery, because of major event or announcement (mergers, acquisition, open offers, de-listings, debt-restructurings, credit-rating downgrades etc.) last day. Fifty constituent stocks of the NIFTY-50 index are part of pre-open trade session. This fifteen minutes session is further broken as 8+4+3. First 8 minutes determine the prices at which the trading would happen. In this session investors can place, modify, or cancel their orders. Orders are not accepted after these eight minutes. In second 4 minutes, orders are matched, followed by price discovery and trade confirmation. A buffer period of last 3 minutes is kept to transfer the state from pre-market session to normal market session. Continuous trading session commences after pre-open session. Block deals takes place during first 35 minutes of this session.

NSE has under-gone major capital market reforms during the last decade i.e. 2000 to 2010. This development includes changes in regulatory framework, introduction of new products and improved market-microstructure. Market microstructure is further improved by employing automated trading system, Application Supported by Blocked Amount (ASBA), dematerialization and reduction in settlement cycle from T+5 to T+2. New products such as derivative trading underlying indices, individual stocks, interest rates and currency; index and gold based ETF and ETF on international indices are introduced. Mini derivative contacts are also launched on Nifty to enhance liquidity and retail participation.

4. Data and Summary Statistics

The time period of the study is four months *i.e.* from 1st April 2015 to 30th July 2015. The study analyses Trade and Quote (TAQ) data of around 113.40 million intraday transactions, comprising of 50 Midcap stocks on 84 trading days. High frequency proxies based on Chordia, Roll and Subrahmanyam (2000, 2001) are employed in the study.

Table 1

Liquidity Proxy Definitions

Liquidity Proxy	Acronym	Definition	Units
Quoted Spread	QS	$QS = \|P_A - P_B\|$	Rupees
Proportional Quoted Spread	PQS	$PQS = \frac{(P_A - P_B)}{P_M}$	Unit less
Effective Spread	ES	$ES = 2 * D_t * \|P_t - P_t^M\|$	Rupees
Depth	DEP	$DEP = \frac{(Q_A + Q_B)}{2}$	Shares
Rupee Depth	RDEP	$RDEP = \frac{(P_A Q_A + P_B Q_B)}{2}$	Rupees

Notes: This table presents definition of liquidity proxies. P_A is the ask price, P_B is the bid price, Q_A, Q_B are the number of shares corresponding to Ask and Bid prices respectively. P_M denotes the mid-price which is calculated as: $P_M = \frac{(P_A + P_B)}{2}$. D_t is 1 (-1) if trade$_t$ was buy (sell) {Boehmer (2005), Bessembinder (2003)}. P_t denotes the last traded price before time *t*. The effective spread is devised to measure actual trading costs.

Liquidity proxies are computed over each transaction where trade and quote data is available. The study aggregates the tick data for each liquidity measures over half an hour{Yang and Choe (2010)}as against other studies which averages liquidity measures over daily basis{Chordia *et al.* (2000);Fabre and Frino (2004); Narayan *et al.* (2015)}. As suggested by Chordia *et al.* (2000), averaging is done in order to achieve greater synchronicity and smooth out intraday effects.

Descriptive statistics for liquidity proxies are reported in Table 2 (Panel A). The table reports that, the standard deviation of all five liquidity proxies is more than the mean values. This shows that the distribution is non-symmetric with a long tail. The medians of all liquidity proxies are smaller than the means indicating distribution to be right skewed. Also, coefficient of variation (CV) is highest in QS and ES proxy. This demonstrates that these proxies have high volatility compared with other proxy measures of liquidity. RDEP and PQS have low volatility.

Table 2 (Panel B) reports correlations between liquidity proxy pairs. The correlation between depth measures (DEP, RDEP) and spread measures (QS) are marginally negative. The lowest correlation is found between QS and RDEP with a value of -0.0135. On other hand, the highest correlation is found between DEP and RDEP with a value of 0.8861.These correlation ranges are similar to other studies. Narayan *et al*. (2015) reported correlation range of -0.0086 to .3825. Pukthuanthong-Le and Visaltanachoti (2009) reported correlation range of -0.03 to 0.98.

Table: 2

Panel A: Summary Statistics of five Liquidity Proxies

Proxy	**Mean**	**Standard deviation**	**CV**	**Median**	**Skewness**	**Kurtosis**
QS	0.880	6.380	7.270	0.120	57.830	5957.840
PQS	0.001	0.001	1.410	0.001	9.570	189.690
ES	1.310	8.880	6.750	0.210	31.400	1568.960
DEP	11325.120	51005.230	4.500	783.210	17.340	672.940
R_DEP	348154.760	657572.000	1.890	183557.950	8.830	151.030

Panel B: Correlation Matrix of five Liquidity Proxies

Proxy	**QS**	**PQS**	**ES**	**DEP**
PQS	0.176			
ES	0.708	0.082		
DEP	-0.029	0.201	-0.031	
RDEP	-0.014	0.178	0.001	0.886

Notes: This table reports cross-sectional statistics and correlations among five liquidity proxies computed from individual stock. The proxies are calculated for every transaction and then averaged within thirty minutes to get one value for each interval. By enforcing Chordia et al. (2000) filters makes the dataset free from problems associated with trading units, trading characteristics, influence of the minimum tick size and infrequently traded stocks. The data set for the study contains stocks from Midcap. If a stock is trading for less than 120 days in a year, then it is excluded from the study.

Table 3 presents half an hour daily variations of liquidity proxies. Proportional changes in spread proxies are higher than that of United States (Chordia *et al.*2000) and Australia (Fabre and Fino 2004) while smaller than that of China (Narayan *et al.*2015).Proportional changes in depth proxies are smaller than that of United States (Chordia *et al.*2000), China (Narayan *et al.* 2015)and Australia (Fabre and Fino, 2004). Comparing with Chordia *et al*. (2000) variations in spread proxies is almost two times larger on NSE Midcap than on NYSE.

Table 3

Summary Statistics: Proportional Changes in Liquidity Variables

Proxy Absolute	Mean	Median	Standard Deviation
|DQS|	0.5175	0.1404	2.5017
|DPQS|	0.5925	0.1512	6.0098
|DES|	0.4427	0.1625	2.3613
|DDEP|	0.7491	0.3057	9.6201
|DRDEP|	0.7501	0.3062	9.6804

Notes: This table reports cross-sectional statistics for percentage change in liquidity proxies (D) for 50 Mid-cap stocks from April to July, 2015. The percentage change in liquidity proxy (L) is computed as $DL_t = (L_t - L_{t-1})/L_{t-1}$ for time *t*.

5. Empirical Design and Evidence

5.1 Market-wide Commonality

To explain the empirical relationship between stock liquidity and market liquidity, the study followed Chordia *et al.* (2000) measures to capture contemporaneous changes in market liquidity as well as lead and leg changes in market liquidity through the following model.

$$DL_{j,t} = \alpha_j + \beta_1 DL_{M,t} + \beta_2 DL_{M,t-1} + \beta_3 DL_{M,t+1} + \varepsilon_{j,t} \quad (1)$$

Where $DL_{j,t}$ is the percentage change (D), for stock *j* from time *t-1* to *t* in the liquidity variable L. $DL_{M,t}$ is the concurrent change in market liquidity which is computed by taking average of the same liquidity measure across stocks excluding stock j. $DL_{M,t+1}$and$DL_{M,t-1}$ are lead and leg of the changes in market liquidity. The above equation includes market squared return as an additional regressor to control possible spurious dependence between returns and bid-ask measures. To empirically tests the presence of commonality in liquidity, the mean sum of concurrent (β_1), lead (β_2) and the lag (β_3) coefficients should be significantly different from zero. The study presents results (Table 4) for value-weighted market liquidity variables on the basis of equation (1).

Table 4 reports that for the percentage change in spread measures *viz.* DQS, DPQS and DES, the average concurrent coefficients are 0.6354, 1.0112 and 0.7530 respectively. Also, over more than 90 percent of the coefficients are significant at 5 percent levels. The results are more pervasive than Chordia *et al.* (2000) and Narayan *et al.* (2015), where over 30 and 80 percent of coefficients are significant at the same level. Mean OLS estimates based on concurrent percentage change in depth proxies, DDEP and DRDEP, are 1.2443 and 0.8344 respectively.

Chordia *et al.* (2000) and Narayan *et al.* (2015) reported coefficients of 1.373 and 73.97 respectively on the DDEP variable. Our study reports that 90 percent of the coefficients on DDEP variable are significant at 5 percent levels. These results are also more pervasive than Chordia *et al.* (2000) and Narayan *et al.* (2015), where 30 and 88 percent of coefficients are significant at the same level.

Although the lead and lag coefficients are positive and negative in case of various liquidity variables, they are small in magnitude as compared with their concurrent counterparts. Similar findings have been reported in USA by Chordia *et al.* (2000), Thailand by Pukthuanthong-Le and Visaltanachoti (2009) and China by Narayan *et al.* (2015). The most significant effects are for lead market liquidity terms on DPQS and DES, where over 29 percent of the coefficients are significant at 5 percent levels.

The sum of concurrent, lead and lag coefficients lies in the range of -1.59 to 0.59. The result reports that the *t*-statistic corresponding to sum is highly significant for the stocks listed on NSE. The computation of *t*-statistic assumes that estimation errors in slope coefficients are independent across regressions (Chordia *et al.*, 2000). The mean adjusted R^2 is less than 2 percent {Chordia *et al.* (2000); Pukthuanthong-Le and Visaltanachoti (2009); Narayan *et al.* (2015); Tayah *et al.* (2015)}. This implies that the individual regressions do not carry impressive explanatory power. Mean Durbin-Watson (DW) statistic has a value of around 2, meaning that the model does not suffers from autocorrelation.

Overall, the results provide strong evidence for the existence of market-wide commonality in liquidity for the Midcap stocks listed on NSE, India. The results of the study corroborate with the existence of strong systematic liquidity phenomenon in the Indian context.

Table 4

Market-wide Commonality in Liquidity

	DQS	DPQS	DES	DDEP	DRDEP
Concurrent (β_1)	0.6354	1.0112	0.7530	1.2443	0.8344
t-statistic	11.5954	10.9727	7.7836	6.6600	3.1704
Median	0.6262	1.0719	0.5522	1.0617	0.5801
Percentage+	100%	98%	100%	98%	82%
Percentage+ significant (5%)	96%	89%	96%	92%	44%
Percentage+ significant (10%)	96%	91%	96%	96%	50%
Lead (β_3)	0.0015	-0.0228	0.0931	-0.8723	-2.3026
t-statistic	0.2839	-0.2311	5.8760	-4.4748	-5.5927
Median	0.0016	0.0689	0.0626	-0.6470	-1.8202
Percentage+	56%	81%	98%	4%	2%
Percentage+ significant (5%)	2%	36%	29%	2%	0%
Percentage+ significant (10%)	4%	43%	42%	4%	2%
Lag (β_2)	-0.0395	0.0068	-0.0229	0.3679	-0.1314
t-statistic	-5.7009	0.3268	-4.1161	3.9448	-1.3759
Median	-0.0424	-0.0155	-0.0301	0.2483	-0.2040
Percentage+	6%	26%	19%	80%	24%
Percentage+ significant (5%)	0%	2%	2%	30%	2%
Percentage+ significant (10%)	0%	2%	2%	40%	4%
Sum	0.5974	0.9952	0.8232	0.7400	-1.5996
t-statistic	11.3170	8.4173	7.6345	2.2567	-3.9252
Median	0.5942	0.9589	0.5534	0.6236	-1.4984
p-value	0.0000	0.0000	0.0000	0.0000	0.0000
Wald test Chi square (Mean)	63.1662	97.9926	65.8112	4.0976	7.0536
Percentage+ significant (5%)	94%	89%	100%	36%	60%
Percentage+ significant (10%)	96%	94%	100%	46%	68%
Adjusted R square Mean	0.1957	0.2453	0.1787	0.0697	0.0616
Adjusted R square Median	0.2062	0.2628	0.1724	0.0575	0.0567
DW Mean	2.0351	2.1179	2.2014	2.1286	2.1092
DW Median	2.0221	2.1056	2.2163	2.1125	2.0790

Notes: This table presents summary of estimates of equation (1). Cross-sectional mean, t-statistic, median, percentage of positive coefficients, percentage of positive and significant coefficients (5% and 10%) of Concurrent, Lead and Lag coefficients. The estimate shows coefficients of percentage changes in individual stock liquidity regressed against percentage changes in value weighted market liquidity for Mid-cap stocks on NSE between the studied period. DW represents Durbin Watson cross-sectional statistics.

5.2 Market-wide and Industry-wide Commonality

The study followed Chordia *et al*. (2001) study on US markets, Pukthuanthong-Le and Visaltanachoti (2009) study on Thailand to examine the strength of market-wide and industry-wide commonality through the following model.

$$DL_{i,t} = \alpha + \beta_1 DL_{M,t} + \beta_2 DL_{M,t-1} + \beta_3 DL_{M,t+1} + \lambda_1 DL_{Ind,t} + \lambda_2 DL_{Ind,t-1} + \lambda_3 DL_{Ind,t+1} + \varepsilon_{i,t} \qquad (2)$$

where, $DL_{i,t}$ is the percentage change of individual stock liquidity. $DL_{M,t}$ is the percentage change of market liquidity and $DL_{Ind,t}$ is the percentage change of industry liquidity. Stock *j* is excluded from the computation of cross-sectional average to derive market and industry liquidity measure. The study employed square market return as an additional regressor. However, the results are not affected if the additional regressors are used. To test the relative strength of market and industry commonality, the mean of the coefficient corresponding to concurrent percentage change in market liquidity and industry liquidity should be significantly different from zero. Also, the sum of concurrent, lead and the lag coefficients of market and industry should be significantly different from zero.

The results reported in Table 5 provide evidences that change in market-wide liquidity and change in industry-wide liquidity explain the change in individual liquidity proxies. The coefficient of market liquidity (concurrent change) and coefficient of industry liquidity (concurrent change) are statistically different from zero. These results are consistent with the findings of Chordia *et al*. (2000), Pukthuanthong-Le and Visaltanachoti (2009), Narayan *et al*. (2015), but inconsistent with the findings of Galariotis and Giouvris (2007), and Tayah *et al*. (2015). For QS, PQS and DEP liquidity measures, the cross-sectional mean of concurrent coefficients on market liquidity is greater than industry liquidity. However, for the ES and RDEP liquidity measures, the cross-sectional mean of concurrent coefficients on industry liquidity is greater than market liquidity. Sum of coefficients on market liquidity is greater than industry liquidity in case of QS and PQS proxy only. In two out of five proxies (ES and RDEP), the concurrent industry liquidity *beta* is greater than market liquidity *beta*. The proportion of positively significant concurrent terms on market liquidity (Table 4) becomes smaller after controlling for the industry effect (Table 5). This additionally shows greater industry effects {Narayan *et al.*,2015}. On the basis of the results, it is difficult to conclude about the relative importance of industry-wide liquidity and market wide liquidity in explaining individual stock liquidity.

The study found both market and Industry individually explains the variations in individual stock liquidity. Individual stock trading volume and spread though decide its level of liquidity of stocks, the turnover of market and industry also influence the level of liquidity. The study concludes that systematic factors in the forms of market and industry explain individual stock liquidity on NSE.

Table5

Industry-wide and Market-wide Commonality in Liquidity

	DQS		DPQS		DES		DDEP		DRDEP	
	Market	Industry	Market	Industry	Market	Industry	Market	Industry	Market	Industry
Concurrent (β_1/ λ_1)	0.5078	0.2225	0.8212	-0.0036	0.3868	0.4927	1.2178	0.1951	0.3754	0.5340
t-statistic	5.6947	2.7216	6.5858	-0.0247	5.1550	2.6989	4.5682	1.0215	1.1552	4.3667
Median	0.3224	0.2988	0.6630	0.1161	0.3150	0.2422	0.8406	0.1908	0.1885	0.4379
Percentage+	94%	84%	92%	63%	89%	79%	90%	64%	58%	82%
Percentage + Significant (5%)	84%	80%	73%	40%	79%	60%	60%	42%	14%	42%
Percentage + Significant (10%)	88%	82%	75%	40%	81%	62%	68%	44%	18%	50%
Lead (β_3/ λ_3)	0.0174	-0.0093	0.5215	-0.0558	0.0862	0.0239	-1.0892	0.0821	-2.2500	0.0130
t-statistic	1.8041	-1.2053	0.4696	-0.0673	3.8139	0.9485	-4.8252	0.5785	-5.7420	0.1811
Median	0.0047	0.0012	0.0425	0.0183	0.0593	0.0081	-0.7892	-0.1023	-2.0017	-0.0129
Percentage+	72%	54%	71%	63%	81%	62%	2%	36%	2%	48%
Percentage + Significant (5%)	2%	2%	10%	8%	19%	9%	2%	18%	0%	6%
Percentage + Significant (10%)	4%	2%	17%	8%	23%	9%	2%	20%	0%	10%
Lag (β_2/ λ_2)	-0.0299	0.0139	-0.2043	0.1145	-0.0631	0.0286	0.2677	0.1373	-0.4251	0.2660
t-statistic	-4.6652	0.9449	-1.0367	1.2367	-3.3511	1.7296	2.7353	2.5467	-3.6054	3.4062
Median	-0.0189	-0.0050	-0.0197	0.0058	-0.0213	-0.0142	0.2079	0.0790	-0.3667	0.1346
Percentage+	18%	40%	35%	52%	28%	43%	78%	68%	22%	70%
Percentage + Significant (5%)	0%	2%	0%	0%	2%	6%	14%	8%	0%	18%
Percentage + Significant (10%)	0%	2%	0%	0%	2%	6%	24%	14%	2%	24%
Sum	0.4953	0.2272	1.1384	0.0552	0.4099	0.5452	0.3963	0.4145	-2.2997	0.8130
t-statistic	5.3420	2.7008	1.3195	0.0672	4.8869	2.6259	0.9815	1.4601	-5.4782	4.1800
Median	0.2992	0.3204	0.8353	0.1402	0.4111	0.2462	0.2078	0.3826	-1.9191	0.4958

	DQS		DPQS		DES		DDEP		DRDEP	
p-value	0.0000	0.0004	0.0000	0.0465	0.0000	0.0000	0.2669	0.0162	0.0000	0.0000
Mean Wald test Chi square	20.5354	30.8193	18.4276	6.9044	19.9116	21.2271	3.1821	4.2617	8.9356	4.6375
Percentage + Significant (5%)	80%	86%	71%	48%	81%	55%	18%	30%	64%	36%
Percentage + Significant (10%)	82%	90%	75%	54%	89%	60%	32%	38%	66%	42%
Adjusted R square Mean	0.2608		0.2518		0.2292		0.0821		0.0714	
Adjusted R square Median	0.2707		0.2640		0.2133		0.0729		0.0650	
DW Mean	2.0302		2.1185		2.2036		2.1226		2.1098	
DW Median	2.0188		2.0996		2.2037		2.0958		2.0861	

Notes: This table presents summary of estimates of equation (2). Cross-sectional mean, t-statistic, median, percentage of positive coefficients, percentage of positive and significant coefficients (5% and 10%) of Concurrent, Lead and Lag coefficients. The estimate shows coefficients of percentage changes in individual stock liquidity regressed against percentage changes in value weighted market liquidity and percentage changes in value weighted industry liquidity for Mid-cap stocks on NSE between the sample period. DW represents Durbin Watson cross-sectional statistics.

5.3 Industry-wide Commonality

Section 5.2 revealed the diminishing role of market liquidity when modelled together with industry liquidity. This implies that industry-wide liquidity is important for Indian stock exchange. Chordia *et al*. (2000) opined that industry specific information creates information asymmetry within that industry. This could affect the degree of responsiveness of the change in stock liquidity to the changes in industry-wide liquidity.

To investigate whether industry specific liquidity is homogeneous or heterogeneous in explaining the individual stock liquidity, the study empirically tested the model employed by Narayan *et al.* (2015). The model (3) is estimated for value-weighted industry liquidity variables ($DL_{Ind,t}$, $DL_{Ind,t-1}$, $DL_{Ind,t+1}$). The study employed square industry return as additional regressors.

$$\mathrm{DL_{j,t}} = \alpha_{\mathrm{j}} + \lambda_1 \mathrm{DL_{Ind\,,t}} + \lambda_2 \mathrm{DL_{Ind\,,t-1}} + \lambda_3 \mathrm{DL_{Ind\,,t+1}} + \varepsilon_{\mathrm{j,t}} \qquad (3)$$

Table 6 to Table 9, presents industry-wide commonality results for Consumer Goods, Financial Services, Manufacturing and Infrastructure sectors. The results common to all industries shows that proportion of positive and significant concurrent terms is higher on QS, PQS and ES proxies and lower on DEP and RDEP proxies. Industry wide commonality is stronger in Manufacturing Sector as the concurrent terms are positive and significant around 63-100 percent of the time. Industry-wide commonality is found weaker on depth based measures (DEP and RDEP). Industry-wide commonality is found stronger on spread based measures (QS and PQS). Strong commonality is found in Manufacturing Sector where 100 percent of the stocks are found to be significant at 5 percent on all spread based measures. This could be attributed to Manufacturing Purchasing Managers Index (PMI) which is found to be above 51 (April-July' 2015) indicating expansion of the sector. Narayan *et al*. (2015) also reported strong commonality for the Industrial manufacturing sector in China, as concurrent terms are significant and positive in 81-87 percent of the stocks.

In terms of the overall industry-wide commonality, Wald test reported that the proportion of significant sum is highest in Financial Services sector (36-100 percent) and lowest in Infrastructure sector (11- 89 percent) at 10% level of significance.

Table 6

Industry-wide Commonality for Consumer Goods Sector

	DQS	DPQS	DES	DDEP	DRDEP
Concurrent (λ_1)	0.6477	0.8475	0.6265	0.3456	0.2314
t-statistic	5.1229	4.9445	8.2386	2.6400	2.1112
Median	0.4899	0.6734	0.6061	0.3569	0.2360
Percentage+	100%	100%	100%	80%	80%
Percentage + significant (5%)	100%	90%	100%	40%	30%
Percentage + significant (10%)	100%	90%	100%	60%	30%
Lead (λ_3)	-0.0037	0.0039	0.0268	-0.5546	-0.4999
t-statistic	-0.2806	0.0844	4.5587	-6.3014	-4.4636
Median	0.0012	0.0488	0.0253	-0.4602	-0.4210
Percentage+	50%	70%	90%	0%	0%
Percentage + significant (5%)	0%	20%	0%	0%	0%
Percentage + significant (10%)	20%	30%	10%	0%	0%
Lag (λ_2)	-0.0175	-0.0381	-0.0282	-0.1157	-0.1580
t-statistic	-3.8996	-1.8974	-2.2723	-1.6955	-3.2281
Median	-0.0174	-0.0238	-0.0310	-0.0884	-0.1131
Percentage+	10%	20%	20%	40%	20%
Percentage + significant (5%)	0%	0%	10%	0%	0%
Percentage + significant (10%)	0%	0%	10%	0%	0%
Sum	0.6265	0.8133	0.6252	-0.3247	-0.4265
t-statistic	4.9957	3.8368	9.0216	-2.2776	-3.9738
Median	0.4925	0.6627	0.5997	-0.3085	-0.4479
p-value	0.0020	0.0039	0.0020	0.0645	0.0059
Mean Wald test Chi square	112.3939	116.1736	66.7836	1.2561	1.2403
Percentage + significant (5%)	100%	90%	90%	10%	0%
Percentage + significant (10%)	100%	90%	90%	10%	20%
Adjusted R square Mean	0.2970	0.2803	0.1923	0.0154	0.0106
Adjusted R square Median	0.2760	0.2907	0.1927	0.0168	0.0103
DW Mean	2.0411	2.1240	2.1630	2.1352	2.1345
DW Median	2.0322	2.1056	2.1825	2.1389	2.1465

Notes: This table presents summary of estimates of equation (3) for Consumer Goods Sector. Cross sectional mean, t-statistic, median, percentage of positive coefficients, percentage of positive and significant coefficients (5% and 10%) of Concurrent, Lead and Lag coefficients are reported. The estimate shows coefficients of percentage changes in individual stock liquidity regressed against percentage changes in value weighted Consumer Goods Sector liquidity for the sample period. DW represents Durbin Watson cross sectional statistics.

Table 7

Industry-wide Commonality for Financial Services Sector

	DQS	DPQS	DES	DDEP	DRDEP
Concurrent (λ_1)	0.6330	0.2191	0.7497	1.2998	1.6051
t-statistic	6.7966	0.3757	8.4671	3.6364	5.0571
Median	0.5543	0.6234	0.7885	1.2495	1.5132
Percentage+	100%	93%	100%	93%	100%
Percentage + significant (5%)	100%	93%	100%	86%	79%
Percentage + significant (10%)	100%	93%	100%	86%	79%
Lead (λ_3)	-0.0113	0.6933	0.0707	-1.0611	-1.4561
t-statistic	-2.2748	0.9808	4.2943	-2.2005	-2.6189
Median	-0.0058	0.0318	0.0681	-0.5666	-0.8274
Percentage+	36%	71%	86%	0%	0%
Percentage + significant (5%)	0%	14%	14%	0%	0%
Percentage + significant (10%)	0%	21%	43%	0%	0%
Lag (λ_2)	-0.0173	-0.3048	-0.0416	0.2655	-0.0587
t-statistic	-5.9912	-1.0910	-5.1173	1.6403	-0.2492
Median	-0.0187	-0.0268	-0.0423	0.2325	0.2016
Percentage+	7%	14%	14%	71%	64%
Percentage + significant (5%)	0%	0%	0%	21%	0%
Percentage + significant (10%)	0%	0%	0%	29%	0%
Sum	0.6043	0.6076	0.7788	0.5041	0.0903
t-statistic	6.7538	3.1741	8.2427	0.5748	0.1642
Median	0.5045	0.6235	0.7989	0.9450	0.7712
p-value	0.0001	0.0107	0.0001	0.0245	0.1726
Mean Wald test Chi square	90.0530	70.7060	79.5304	6.8714	2.6911
Percentage + significant (5%)	100%	93%	100%	57%	36%
Percentage + significant (10%)	100%	93%	100%	64%	50%
Adjusted R square Mean	0.2504	0.2075	0.2204	0.0620	0.0604
Adjusted R square Median	0.2408	0.1691	0.2160	0.0652	0.0684
DW Mean	2.0226	2.1257	2.1908	2.1027	2.0812
DW Median	2.0224	2.1524	2.1850	2.0977	2.0709

Notes: This table presents summary of estimates of equation (3) for Financial Services Sector. Cross sectional mean, t-statistic, median, percentage of positive coefficients, percentage of positive and significant coefficients (5% and 10%) of Concurrent, Lead and Lag coefficients are reported. The estimate shows coefficients of percentage changes in individual stock liquidity regressed against percentage changes in value weighted Consumer Goods Sector liquidity for the sample period. DW represents Durbin Watson cross sectional statistics.

Table 8

Industry-wide Commonality for Manufacturing Sector

	DQS	DPQS	DES	DDEP	DRDEP
Concurrent (λ_1)	0.4484	0.8127	0.4168	0.6485	0.5731
t-statistic	6.4029	5.4975	12.8024	4.6376	3.4809
Median	0.4546	0.7971	0.4119	0.5634	0.5355
Percentage+	100%	100%	100%	100%	88%
Percentage + significant (5%)	100%	100%	100%	88%	63%
Percentage + significant (10%)	100%	100%	100%	88%	63%
Lead (λ_3)	-0.0077	0.0193	0.0362	-0.2363	-0.5716
t-statistic	-1.3482	0.7483	4.4992	-3.3980	-3.6400
Median	-0.0079	-0.0040	0.0257	-0.1567	-0.4234
Percentage+	50%	50%	100%	0%	0%
Percentage + significant (5%)	0%	17%	14%	0%	0%
Percentage + significant (10%)	0%	17%	14%	0%	0%
Lag (λ_2)	-0.0402	-0.0437	-0.0067	-0.0519	-0.1924
t-statistic	-5.6427	-4.3137	-1.0011	-1.7716	-3.7750
Median	-0.0425	-0.0501	0.0030	-0.0682	-0.1229
Percentage+	0%	0%	57%	38%	0%
Percentage + significant (5%)	0%	0%	0%	0%	0%
Percentage + significant (10%)	0%	0%	0%	0%	0%
Sum	0.4005	0.7884	0.4463	0.3603	-0.1909
t-statistic	6.0769	4.8932	13.4907	2.2423	-0.9245
Median	0.3799	0.7349	0.4399	0.2415	0.1070
p-value	0.0078	0.0313	0.0156	0.0547	0.8438
Mean Wald test Chi square	37.9799	75.2785	42.0043	6.5697	2.0132
Percentage + significant (5%)	100%	100%	100%	38%	25%
Percentage + significant (10%)	100%	100%	100%	38%	25%
Adjusted R square Mean	0.1339	0.2328	0.1151	0.0623	0.0337
Adjusted R square Median	0.1439	0.2334	0.1153	0.0542	0.0371
DW Mean	2.0275	2.1196	2.2160	2.1801	2.1887
DW Median	2.0278	2.1092	2.2124	2.1464	2.1614

Notes: This table presents summary of estimates of equation (3) for Manufacturing Sector. Cross sectional mean, t-statistic, median, percentage of positive coefficients, percentage of positive and significant coefficients (5% and 10%) of Concurrent, Lead and Lag coefficients are reported. The estimate shows coefficients of percentage changes in individual stock liquidity regressed against percentage changes in value weighted Consumer Goods Sector liquidity for the sample period. DW represents Durbin Watson cross sectional statistics.

Table 9

Industry-wide Commonality for Infrastructure Sector

	DQS	DPQS	DES	DDEP	DRDEP
Concurrent (λ_1)	0.5152	0.9778	1.0351	0.8183	0.6963
t-statistic	6.0139	5.1987	2.6626	10.9255	10.5722
Median	0.4519	0.8080	0.4125	0.8627	0.7447
Percentage+	100%	94%	100%	100%	94%
Percentage + significant (5%)	94%	67%	88%	94%	89%
Percentage + significant (10%)	94%	72%	88%	94%	89%
Lead (λ_3)	-0.0058	0.0330	0.1498	-0.3181	-0.7414
t-statistic	-0.7134	0.2433	3.3501	-3.1728	-5.2179
Median	0.0021	0.0790	0.0799	-0.4092	-0.8380
Percentage+	61%	72%	88%	11%	11%
Percentage + significant (5%)	0%	33%	47%	6%	0%
Percentage + significant (10%)	0%	33%	59%	11%	0%
Lag (λ_2)	0.0036	0.0812	-0.0204	0.1037	0.1414
t-statistic	0.1090	1.2128	-1.3834	2.4408	1.7741
Median	-0.0248	-0.0169	-0.0337	0.0824	0.0593
Percentage+	17%	33%	24%	67%	56%
Percentage + significant (5%)	6%	6%	0%	11%	22%
Percentage + significant (10%)	6%	6%	0%	22%	28%
Sum	0.5130	1.0919	1.1646	0.6040	0.0963
t-statistic	6.1569	6.2571	2.6674	6.2507	0.6210
Median	0.5551	0.9070	0.4204	0.5898	-0.0125
p-value	0.0000	0.0000	0.0000	0.0000	0.9323
Mean Wald test Chi square	60.2814	70.3962	69.5727	5.2252	2.1988
Percentage + significant (5%)	89%	72%	94%	61%	11%
Percentage + significant (10%)	89%	83%	94%	61%	11%
Adjusted R square Mean	0.1817	0.1761	0.1600	0.0684	0.0530
Adjusted R square Median	0.1783	0.1430	0.1447	0.0669	0.0505
DW Mean	2.0497	2.1095	2.2025	2.1680	2.1707
DW Median	2.0241	2.0873	2.2288	2.1447	2.1411

Notes: This table presents summary of estimates of equation (3) for Infrastructure Sector. Cross sectional mean, t-statistic, median, percentage of positive coefficients, percentage of positive and significant coefficients (5% and 10%) of Concurrent, Lead and Lag coefficients are reported. The estimate shows coefficients of percentage changes in individual stock liquidity regressed against percentage changes in value weighted Consumer Goods Sector liquidity for the sample period. DW represents Durbin Watson cross sectional statistics.

Section 5.3 concludes that stocks belonging to Infrastructure sector are less sensitive to the liquidity of their sector. However, the stocks belonging to Financial Services Sector are more sensitive to the liquidity of their sector.

5.4 Commonality in up and down markets

As discussed in literature {Sujoto, Kalev and Faff (2004); Kempf and Mayston (2005)}, commonality vary in up and down phases of the markets. The study has empirically examined the same using modified model of Narayan *et al*. (2015) model in Indian context.

$$DL_{j,t} = \alpha_j^u D_u + \alpha_j^d D_d + \beta_j^u D_u D_{M,t} + \beta_j^d D_d DL_{M,t} + \gamma_j DL_{j,t-1} + \epsilon_{j,t} \qquad (4)$$

The study uses both intercept and slope dummy (D_u and D_d) to study impact of upward and downward market movements on short run liquidity beta. Fabozzi and Francis (1977), Kim and Zumwalt (1979) and Chen (1982) have defined bull market in which returns are greater than a certain threshold value. This study defines the market to be up or down on the basis of excess market return over 10 years yield of government securities (G-Sec rate) rate. The dummy variables for the same are defined as:

$$R_m - R_f > 0, Bull\ Market, D_u = 1, D_d = 0$$

$$R_m - R_f \leq 0, Bear\ Market, D_u = 0, D_d = 1$$

The study investigates time series determinants of commonality in liquidity at the individual security level. Market-wide factors are separated from firm specific factors in order to study how different factors affect individual liquidity.

The results reported in Table 10 reports asymmetry of commonality in up and down markets. On the up market, cross sectional mean of the up market slope dummy (β_j^u) is positive and significant in over 66 percent of the stocks. On the down market, cross sectional mean of the down market slope dummy (β_j^d) coefficient is positive and significant in over 70 percent of the stocks. This is in contrast with the results of Narayan *et al*. (2015), where β_j^d on Depth and Turnover is found to be positive and significant in over 10 percent of stocks.

On the NSE, the highest cross sectional mean of β_u is 1.7297 (based on DRDEP) and the lowest is 0.4909 (based on DPQS). While the highest cross sectional mean of β_d is 1.1737 (based on DDEP) and the lowest is 0.5119. The slope dummy coefficient in up markets is found stronger than on down markets on all liquidity measures except PQS. This implies that the herd behaviour of Indian investors is more pronounced in up markets compared with the down-markets in the studied period. Strong commonality in up-markets is consistent with the findings of Sujoto, Kalav and Faff (2004), while inconsistent with Kempf and Maytson (2005), Kumar and Shah (2012), and Narayan *et al*. (2015).

Wald test examines the null hypothesis regarding the symmetry of slope dummy coefficient in up markets (β_u) and down markets (β_d). At 10 percent level of significance, over 30 percent of the stocks rejects the null hypothesis. QS proxy reports highest asymmetry in commonality on up and down markets with over 64 percent of the stocks rejecting the null hypothesis at 10 percent level of significance. The findings provides evidence that commonality varies between up and down markets in India.

Table 10

Commonality Asymmetry on Up and Down markets

	DQS	DPQS	DES	DDEP	DRDEP
β_u(Mean)	0.7730	0.4909	1.1416	1.6821	1.7297
t-statistic	9.4542	0.9786	4.6222	4.8255	4.2922
β_u (Median)	0.7049	1.0009	0.5743	1.2720	1.2240
Percentage +	100%	98%	100%	98%	98%
Percentage + significant (5%)	94%	87%	96%	84%	66%
Percentage + significant (10%)	96%	87%	96%	88%	68%
β_d(Mean)	0.5805	0.9850	0.5119	1.1737	1.6320
t-statistic	11.7300	10.7095	15.5605	8.1555	3.1683
β_d (Median)	0.5787	0.9054	0.4895	0.9643	1.1511
Percentage +	100%	98%	100%	98%	98%
Percentage + significant (5%)	96%	83%	92%	90%	70%
Percentage + significant (10%)	96%	85%	92%	90%	74%
Wald Test Results					
Mean Chi Square	18.0921	12.6638	35.2407	6.8822	3.2035
Median Chi Square	9.5476	5.3640	5.0257	3.1682	0.7403
Percentage significant (5%)	60%	51%	52%	46%	24%
Percentage significant (10%)	64%	57%	60%	54%	32%
Adjusted R square Mean	0.2282	0.2674	0.2245	0.0867	0.0573
Adjusted R square Median	0.2329	0.2911	0.2248	0.0852	0.0598
DW Mean	1.9784	2.0200	2.0352	2.0341	2.0133
DW Median	1.9769	2.0099	2.0258	2.0296	2.0068

Notes: This table presents summary of estimates of equation (4). The estimates shows the strength of commonality on up and down markets on NSE. The null hypothesis for the Wald test is H_0: $\beta_u = \beta_d$. DW and χ^2 represent Durbin Watson and χ^2 cross-sectional statistics. The table also reports the percentage of stocks that rejects the null hypothesis at 5% and 10% level of significance.

5.5 Long Run Commonality Beta

Short-run liquidity beta (β_1) or the concept of systematic liquidity is explored by Chordia *et al.* (2000). It captures the co-movement of individual stock liquidity with market liquidity in short run. However, the study of Faff *et al.* (2002) captures liquidity co-movements in the long run. Our study followed Faff *et al.* (2002) in order to estimate the long run systematic liquidity beta by augmenting the standard model of Chordia *et al*. (2000) with a lagged dependent variable ($DL_{j,t-1}$).

$$DL_{j,t} = \alpha_j + \beta_{LR}(1-\lambda_j)DL_{M,t} + \lambda_j DL_{j,t-1} + \varepsilon_{j,t} \tag{5}$$

Where β_{LR} is the long run liquidity beta and is related to short run 'impact' coefficient of liquidity beta as,

$$\beta_{LR} = \frac{\beta_1}{(1-\lambda)}, where\ \lambda < 0 \tag{6}$$

The results reported in Table 11 reports summary of long-run beta estimation results. Many emerging markets, like India (NSE), are thinly traded in comparison to the developed markets like USA. At 10 percent level of significance, 90-96 percent of the stocks have positive and significant β_{LR}. Additionally, cross-sectional mean of the long run liquidity beta (β_{LR}) is positive and significant on all measures of liquidity. The highest mean β_{LR} is 1.6180 (RDEP) and the lowest is 0.5823 (QS). These results indicate the presence of long run liquidity commonality in Indian markets.

The coefficient of lagged term (λ_j) should be less than zero is the condition for equation (5.5) to be valid. At 10 percent level of significance, 90-96 percent of the stocks have negative and significant λ_j. These results, confirms the validity of the model in Indian context. Additionally, cross-sectional mean of λ_j is negative and significant on all measures of liquidity. The lowest mean λ_j is -0.0925 (ES) and the highest is -0.0353 (QS). Negative λ_j explains the mean reversal tendency of time-varying liquidity beta.

Consistent reporting of $\lambda_j<0$, indicates that long run commonality is smaller than short run commonality beta. Comparing the coefficients of concurrent term (β_1) in Table 4 and β_{LR} in Table 11, short run commonality beta is stronger on all measures except DEP. Madhavan (2000) explains that short run impact is generated by noise traders while permanent effect is generated by informed traders.

The result indicates that on Indian markets, the commonality is more because of noise traders than because of informed traders. Intra-period (Half an hour) fluctuations of individual stock liquidity (λ_j), is over 3 percent and this carry forward liquidity requirement has long run adjustment in the form of β_{LR} varies above 50 percent.

Table11

Long-Run Liquidity Beta Estimation

	DQS	DPQS	DES	DDEP	DRDEP
$\boldsymbol{\beta_{LR}}$ (Mean)	0.5823	0.9493	0.6539	1.3480	1.6180
t-statistic	11.4124	11.0076	7.8125	7.7693	4.3555
$\boldsymbol{\beta_{LR}}$(Median)	0.5941	0.8815	0.4870	1.1513	1.1892
Percentage +	100%	98%	100%	100%	98%
Percentage + Significant (5%)	96%	89%	96%	94%	88%
Percentage + Significant (10%)	96%	91%	96%	94%	90%
$\boldsymbol{\lambda_j}$ (Mean)	-0.0353	-0.0505	-0.0925	-0.0562	-0.0675
t-statistic	-7.1438	-8.4703	-13.9056	-7.1588	-9.0940
$\boldsymbol{\lambda_j}$(Median)	-0.0305	-0.0468	-0.0949	-0.0411	-0.0561
Percentage -	96%	96%	96%	92%	96%
Percentage - Significant (5%)	10%	46%	83%	36%	40%
Percentage - Significant (10%)	12%	54%	83%	44%	52%
Percentage <1	100%	100%	100%	100%	100%
Adjusted R square Mean	0.2133	0.2543	0.1857	0.0562	0.0293
Adjusted R square Median	0.2144	0.2664	0.1805	0.0515	0.0261
DW Mean	1.9695	2.0198	2.0279	2.0338	2.0143
DW Median	1.9631	2.0093	2.0192	2.0292	2.0079

Notes: This table presents summary of estimates of equation (5). Value weighted cross-sectional means of time-series slope coefficients are reported with the corresponding t-statistics. DW represents Durbin Watson cross-sectional statistics. The table reports that percentage of stocks having long run beta positive and significant at 5% and 10% level of significance.

6. Conclusions

NSE is third biggest stock exchange of Asia and fifth biggest stock exchange of world in terms of transactions (Mukherjee, 2007). India is the growing force in the world economy. Liquidity shocks in Indian stock market not only have relevance in the Asian region but it would inevitably affect the rest of the world. The detailed investigation of commonality on NSE will help in understanding asset pricing (Acharya and Pedersen, 2005; Korajczyk and Sadka, 2008; Pastor and Stambaugh, 2003). The understanding will also help investors to bear commonality risk with greater efficiency and improving investment strategies. As commonality risk is non-diversifiable it is policy concern for securities market regulator (SEBI). Shocks to commonality will have market wide and industry wide effects and hence effect the functioning of Indian stock markets as a whole. Coughenour and Saad (2004) argued that knowing commonality dynamics will aid regulators and exchanges in improving market microstructure.

Our study investigated liquidity commonality hypothesis for a sample of NSE Midcap stocks. Over 50 percent of the stocks have positive and significant concurrent *beta* on all measures of liquidity. This shows that the effect of commonality in liquidity is generally more perseverant on Indian stock market than that reported in other stock markets. The study found the existence of industry-wide commonality and market-wide wide commonality on the Midcap stocks listed on NSE. This further suggests that asymmetric information can be the driver of systematic liquidity on NSE (Chordia *et al.*,2000). The comparative strength of liquidity commonality varies between Industry and Market depending on the measure of liquidity. Portfolio managers face challenges in managing portfolios in up-markets as commonality is found stronger in up phase. Manufacturing sector reports the highest degree of commonality among all the sectors. Systematic liquidity of Midcap stocks moves in asymmetric manner in up and down markets. The study developed long-run commonality *beta* for the Midcap stocks which indicates that liquidity movement in Indian stock market may be influenced by more of noise traders rather than informed traders.

Our study on liquidity commonality shows that the Indian stock market exhibits similarities as well as differences with the other studies on quote-driven and order-driven stock markets. Given the scarcity of relevant theories and literature, further research is warranted in this area of market microstructures. Empirically testing the reasons for commonality, using high frequency data with wide array of stocks could help in building a theoretical market micro-structure model.

Acknowledgement

This work is supported by University Grants Commission (UGC), India under Junior Research Fellowship (JRF). The authors would like to thank members of Doctoral Scrutiny Committee (DSC) at IIT Kharagpur and anonymous reviewers for their helpful comments.

Disclosure statement

No potential conflict of interest was reported by the authors.

References

Acharya, V.V. and Pederson, L.H. (2005). Asset pricing with liquidity risk. *Journal of Financial Economics*, 77, 375–410. doi:10.1016/j.jfineco.2004.06.007.

Admati, A. and Pfleiderer, P. (1988). A theory of intraday patterns: volume and price variability, *Review of Financial Studies*, 1, 3-40. doi:10.1093/rfs/1.1.3

Amihud, Y. and Mendelson, H. (1980). Dealership Market. *Journal of Financial Economics*,8, 31-53. doi:10.1016/0304-405X(80)90020-3

Anderson, R.G., Binner, J.M., Hagströmer, B. and Nilsson, B. (2013). Does commonality in illiquidity matter to investors?, *Federal Reserve Bank of St. Louis Working Paper Series*, 2013-020A.

Bessembinder, H. (2003). Issues in assessing trade execution costs. *Journal of Financial Markets*, 223-257. doi:10.1016/s1386-4181(02)00064-2

Boehmer, E. (2005). Dimensions of execution quality: Recent evidence for US equity markets. *Journal of Financial Economics*, 78 (3), 553-582. doi:10.1016/j.jfineco.2004.11.002

Brockman, P., and Chung, D.Y. (2002). Commonality in Liquidity: Evidence from an order-driven market structure. *The Journal of Financial Research*, 4, 521-539. doi:10.1111/jfir.2002.25.issue-4.

Brockman, P., and Chung, D.Y. (2006). Index inclusion and commonality in liquidity: Evidence from the Stock Exchange of Hong Kong. *International Review of Financial Analysis*, 15, 291-305. doi:10.1016/j.irfa.2005.09.003.

Chen, S. N. (1982). An examination of risk return relationship in bull and bear markets using time varying betas. *Journal of Financial and Quantitative Analysis*, 17(2), 265–286. doi:10.2307/2330850

Choe, H. and Yang, C.-W. (2010), Liquidity Commonality and its Causes: Evidence from the Korean Stock Market. *Asia-Pacific Journal of Financial Studies*, 39: 626–658. doi:10.1111/j.2041-6156.2010.01025.x.

Chordia, T., Roll, R. and Subrahmanyam, A. (2000). Commonality in liquidity. *Journal of Financial Economics*, 56, 3–28. doi:10.1016/S0304-405X(99)00057-4.

Chordia, T., Roll, R. and Subrahmanyam, A. (2001).Market liquidity and Trading Activity. *Journal of Finance*, 56(2), 501–530. doi:10.1111/jofi.2001.56.issue-2.

Coughenour, J.F. and Saad, M.M. (2004). Common market makers and commonality in liquidity. *Journal of Financial Economics*, 73, 37–69. doi:10.1016/j.jfineco.2003.05.006.

Dicle, M.F. and Mukherjee, T.K. (2011). Intraday Trading Patterns and Commonality in Liquidity. Available at SSRN: http://ssrn.com/abstract=1925826. doi:10.2139/ssrn.1925826

Domowitz, I., Hansch, O., and Wang, X. (2005). Liquidity commonality and return co-movement. Journal of Financial Markets, 8, 351-376. doi:10.1016/j.finmar.2005.06.001.

Fabozzi, F.J. and Francis, J.C. (1977).Stability Tests for Alphas and Betas over Bull and Bear Market Conditions. *Journal of Finance*, 32, 1093-1099. doi:10.1111/j.1540-6261.1977.tb03312.x

Faff , R., Kalev, P., and Sujoto, C. (2008). Systematic liquidity in the long run. *Applied Financial Economics Letters*, 4, 187-191. doi:10.1080/17446540701591357

Fabre, J., and Frino, A. (2004). Commonality in liquidity: evidence from the Australian stock exchange. *Accounting and Finance*, 44, 357-368. doi:10.1111/acfi.2004.44.issue-3.

Galariotis, E. C. and Giouvris, E. (2007), Liquidity Commonality in the London Stock Exchange. *Journal of Business Finance & Accounting*, 34, 374–388. doi: 10.1111/j.1468-5957.2006.00664.x

Giouvris, E. (2003). Systematic liquidity and expected returns: evidence from the London Stock Exchange. Working Paper, University of Durham.

Goyal A., Syamala, S.R. and Reddy, V.N. (2014). Commonality in liquidity: An empirical examination of emerging order-driven equity and derivatives market, *Journal of International Financial Markets, Institutions & Money*, 33, 317–334. doi: 10.1016/j.intfin.2014.09.001

Grossman, S. J. and Miller, M. H. (1988), Liquidity and Market Structure. *Journal of Finance*, 43: 617–633. doi:10.1111/j.1540-6261.1988.tb04594.x

Hasbrouck, J., and Seppi, D.J. (2001). Common factors in prices, order flows, and liquidity. *Journal of Financial Economics*, 59, 383–411. doi:10.1016/S0304-405X(00)00091-X

Huberman, G. and Halka, D. (2001), Systematic Liquidity. *Journal of Financial Research*, 24: 161–178. doi: 10.1111/j.1475-6803.2001.tb00763.x

Kempf, A. and Mayston, D. (2005). Commonalities in the liquidity of a limit order book. Available at: http://www.fbv.kit.edu/symposium/10th/papers/Mayston_Kempf%20-%20Commonalities%20in%20the%20Liquidity%20of%20a%20Limit%20Order%20Book.pdf

Kim, M. K. and Zumwalt, J. K. (1979). An analysis of risk in bull and bear markets, *Journal of Financial and Quantitative Analysis*, 14(5), 1015–1025. doi:10.2307/2330303

Korajczyk, R.A., and R. Sadka. 2008. Pricing the Commonality Across Alternative Measures of Liquidity. *Journal of Financial Economics*, 87, 45‑72. doi: 10.1016/j.jfineco.2006.12.003

Krishnan, R. and Mishra, V. (2013). Intraday Liquidity Patterns in Indian Stock Market. *Journal of Asian Economics*, 28, 99–114. doi:10.1016/j.asieco.2013.05.005

Kumar, S. and Shah, A. (2012). Commonality in Liquidity of an Open Electronic Limit Order Book Market. Available at: http://www.igidr.ac.in/conf/money/mfc_08/Commonality%20in%20Liquidity%20of%20an%20Open%20Electronic%20Limit%20Order%20Book%20Market_SantoshKumar.pdf

Kyle, A.S., (1985). Continuous auctions and insider trading. *Econometrica*, 53, 1315-1335. doi:10.2307/1913210

Mukherjee, D. (2007). Comparative Analysis of Indian Stock Market with International Markets. *Great Lakes Herald*, 1(1).

Narayan, P.K. and Zheng, X. (2011).The relationship between liquidity and returns on the Chinese stock market. *Journal of Asian Economies*, 22, 259-266. doi:10.1016/j.asieco.2011.02.005

Narayan, P.K., Zhang, Z. and Zheng X. (2015). Some Hypothesis on Commonality in Liquidity: New Evidence from the Chinese stock Market. *Emerging Markets Finance and Trade*, 51, 5. doi:10.1080/1540496X.2015.1061799

Papachristou, G., Anagnostidis, P., and Thomaidis, N.S. (2016). Liquidity commonality in order-driven trading: evidence from the Athens Stock Exchange, *Applied Economics*, 48:22, 2007-2021. doi: 10.1080/00036846.2015.1111991.

Pastor, L., and R. Stambaugh. (2003). Liquidity Risk and Expected Stock Returns. *Journal of Political Economy*, 111, 642‑685.

Pukthuanthong-Le, K. and Visaltanachoti, N. (2009). Commonality in liquidity: Evidence from the Stock Exchange of Thailand. *Pacific-Basin Finance Journal*, 17, 80–99. doi:10.1016/j.pacfin.2007.12.004.

Sujoto, C., Kalev, P. S. and Faff, R. W. (2004). Commonality in liquidity: Further Australian evidence, *Accounting & Finance*, 44(3), 357–368. doi: 10.1111/j.1467-629x.2004.00117.x.

Sujoto, C., Kalev, P.S. and Faff, R.W. (2005). An examination of commonality in liquidity: new evidence, long-run effects and non-linearities, Working Paper, Department of Accounting and Finance, Monash University.

Tayah, M., Bino, A., Ghunmi, D.A. and Tayem, G. (2015). Liquidity Commonality in an Emerging Market: Evidence from the Amman Stock Exchange, *International Journal of Economics and Finance*, 7, 2. doi:10.5539/ijef.v7n2p203

Appendix I

NSE's Midcap 50 Sample Stocks[i]

Company Name	Industry	Market Cap
Aditya Birla Nuvo Ltd.	Services	216,894.4
Bata India Ltd.	Consumer Goods	70,349.5
Biocon Ltd.	Pharma	93,210.0
Godrej Industries Ltd.	Consumer Goods	120,591.7
Havells India Ltd.	Consumer Goods	190,375.2
Jubilant Foodworks Ltd.	Consumer Goods	96,817.1
McLeod Russel India Ltd.	Consumer Goods	26,329.6
Strides Arcolab Ltd.	Pharma	71,458.3
Sun TV Netwrok Ltd.	Media & Entertainment	179,702.6
Tata Global Beverages Ltd.	Consumer Goods	93,440.0
Allahabad Bank	Financial Services	55,223.4
Andhra Bank	Financial Services	47,493.5
Bank of India	Financial Services	136,804.9
Canara Bank	Financial Services	175,739.6
IDBI Bank Ltd.	Financial Services	116,768.1
IFCI Ltd.	Financial Services	56,841.7
Karnataka Bank Ltd.	Financial Services	24,083.3
L&T Finance Holdings Ltd.	Financial Services	109,238.2
Oriental Bank of Commerce	Financial Services	63,058.2
Power Finance Corporation Ltd.	Financial Services	373,109.5
Reliance Capital Ltd.	Financial Services	109,276.3
SKS Microfinance Ltd.	Financial Services	59,080.3
Syndicate Bank	Financial Services	64,582.0
Union Bank of India	Financial Services	103,028.0
Apollo Tyres Ltd.	Automobile	89,181.1
Ashok Leyland Ltd.	Automobile	210,737.2
Crompton Greaves Ltd.	Industrial Manufacturing	104,102.5
Jain irrigation Systems Ltd.	Industrial Manufacturing	28,509.2
MRF Ltd.	Automobile	167,766.9
Siemens Ltd.	Industrial Manufacturing	503,019.8
Tata Chemicals Ltd.	Chemicals	112,373.0
TVS Motor Company Ltd.	Automobile	121,052.2
CESC Ltd.	Energy	80,826.7
GMR Infrastructure Ltd.	Construction	89,234.2
Hindustan Petroleum Corporation Ltd.	Energy	221,259.1
Hindustan Zinc Ltd.	Metals	691,473.5
India Cements Ltd.	Cement & Cement Products	27,354.3
Indraprastha Gas Ltd.	Energy	57,722.1
IRB Infrastructure Developers Ltd.	Construction	87,335.3
Jaiprakash Associates Ltd.	Cement & Cement Products	60,933.0
JSW Energy Ltd.	Energy	195,658.6

Company Name	Industry	Market Cap
Justdial Ltd.	IT	93,674.3
NHPC Ltd.	Energy	221,966.9
Oracle Financial Services Software Ltd.	IT	275,349.4
Petronet LNG Ltd.	Energy	131,925.0
Reliance Infrastructure Ltd.	Energy	116,346.8
Reliance Power Ltd.	Energy	160,733.7
Steel Authority of India Ltd.	Metals	289,932.5
Unitech Ltd.	Construction	42,357.1
Voltas Ltd.	Construction	93,872.0

[i] The market capitalization (INR 10 million or Crore) of Midcap stocks as on 1st April, 2015.

Appendix II

Largest Stock Exchange in Asia (2016 Rankings)

Rank	Asia Stock Exchange	Total Market Capitalization	% Change (USD)
1	Tokyo Stock Exchange	$5.1 Trillion	3.40%
2	Shanghai Stock Exchange	$4.1 Trillion	9.80%
3	Shenzhen Stock Exchange	$3.2 Trillion	6.10%
4	Hong Kong Stock Exchange	$3.2 Trillion	0.30%
5	National Stock Exchange	$1.5 Trillion	3.00%

Source: Caproasia Institute